

Scalp Microinfusion: A Novel Drug Delivery Technique for Hair Loss Treatment

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Keywords

Scalp microinfusion · MMP[®] · Transdermal drug delivery · Novel drug delivery · Hair loss · Alopecia · Female pattern hair loss · Male pattern hair loss · Alopecia areata · Triamcinolone · Dutasteride · Finasteride · Minoxidil

Abstract

Scalp microinfusion is a promising novel drug delivery technique for hair loss treatment. We discuss the MMP[®] technique and review its possible use in alopecias. MMP[®] technique provides a small amount of drugs delivered homogeneously into the skin combined with micro-needling and can, therefore, provide optimal delivery. However, literature on this technique is limited to a few case reports despite its wide use in some countries. Further studies are needed to standardize protocols.

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Introduction

Many drug delivery approaches for hair loss treatment have been developed in the last few years [1, 2]. MMP[®] (the Portuguese acronym for microinfusion of drugs into the skin) is a novel drug delivery technique that uses mi-

cro-needling of professional tattoo machines to infuse drugs into the skin [1–6]. It was first described in the treatment of keloid and idiopathic guttate hypomelanosis [3].

Recently, this method has gained popularity as adjunct therapy for alopecia, despite limited data. This article aimed to review the literature on the MMP[®] technique, especially for hair loss treatment.

How Does MMP[®] Work?

This technique promotes drug delivery via micro-needling using a professional tattoo machine with appropriate needles and aseptic standards. The drug solutions to be infused are sterile and can be the same ones usually used for mesotherapy. The solution is placed in a container and loaded onto the cartridge by capillarity. The device then conveys it into the skin. The speed and needling depth can be adjusted depending on the condition to be treated and the endpoint [1–6]. Delivery with MMP[®] is not affected by the medication's chemical properties [4, 6].

Is MMP[®] Safe?

To assess MMP[®] safety, Arbache et al. [4] performed an experimental study quantifying the drug delivered by weighing human skin samples before and after MMP[®]

Table 1. Summary of case reports on MMP® technique for hair loss treatment

Author	Subjects/ gender	Alopecia type	Drug delivered	Intervention	Results
Contin, (2016) [5]	Two males	Androgenetic alopecia	Minoxidil 0.5%	4-monthly MMP® sessions with drug infusion versus 3-monthly MMP® sessions without infusion 1.5 mm needle depth	Satisfactory partial response No superiority of MMP® with drug delivery
Barletta and Gasques, (2020) [6]	One male and one female	AA	TA 2.5 mg/mL and 10 mg/mL	4-monthly MMP sessions comparing the two concentrations of TA 1.0 mm needle depth	Promising intervention that stimulated hair growth in the AA patches

AA, alopecia areata; TA, triamcinolone acetonide.



Fig. 1. Clinical image of a patient with female pattern hair loss before treatment (a) showing good outcome after 4-monthly MMP® sessions with minoxidil 0.5% infusion (b).

sessions. They used a Cheyenne® tattoo machine and cartridges containing 27 needles with a diameter of 0.3 mm and tip length of 5.5 mm each. The needling depth was set to 300 microns and the frequency to 120 Hz. A small amount of medication (1,175 µg/cm²) was diffusely scattered in the superficial dermis without an expansive effect. The adjustment of the medication delivery depth showed to increase the safety of this procedure.

MMP® and Hair Loss

Although MMP® has increasingly been used, data on treatments for hair disorders are still limited to a few case reports. Table 1 shows its successful application in androgenetic alopecia and alopecia areata. Figure 1 a-b illustrates a good result after MMP® sessions in a female pattern hair loss patient.

MMP® is well tolerated by most patients. Pain can usually be managed with topical anesthesia, vibration devices, or the pinch technique (in which the scalp is grasped

between the thumb and index finger of the nondominant hand) [7, 8]. In the authors' experience, most cases do not require topical anesthesia.

MMP® is an innovative way to stimulate hair growth as it combines uniform injections of small amounts of drugs into the dermis with micro-needling. Some studies have described depths from 1.0 mm to 1.5 mm depending on the treated condition [5, 6]. It is similar to the micro-needling technique in which the optimal depth is between 1.2 and 1.5 mm using medical devices such as dermaroller® or electric pen [1].

Compared with intralesional injections, MMP® requires less volume, reduces wasting and unwanted local side effects, and improves efficiency due to higher precision [1, 2]. So far, severe side effects have not been reported [6]. Additionally, as it occurs in micro-needling procedures using dermaroller® or electric pen, MMP® enables tissue regeneration through stem cell activation in the bulb and overexpression of hair growth-related

genes such as Wnt3a, Wnt10-b, and vascular endothelial growth factor [5]. In the author's experience, MMP[®] does not cause shaft breakage, sometimes seen with traditional micro-needling devices. The main limitation of MMP[®] is the lack of standardization regarding the procedure execution, such as the number of sessions needed, the optimal interval between sessions, the maintenance protocol, and the ideal blend of drugs to be infused.

Conclusion

MMP[®] is a promising drug delivery technique with the advantage that a small amount of drug can be delivered homogeneously into the skin at the same depth. Even if no medication is infused, the small orifices produced may have an effect similar to other micro-needling devices.

This article provides a review of MMP[®] for the treatment of hair loss – a topic still poorly explored that has progressively gained popularity despite the lack of solid evidence. Further studies are needed to establish this technique's efficacy and develop standardized protocols. MMP[®] appears to be a potentially safe adjuvant therapy for numerous hair disorders, including alopecia areata and androgenetic alopecia.

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Statement of Ethics

The study complies with the internationally accepted standards for research practice and reporting. Ethical approval was not required for this study, following national guidelines.

Conflict of Interest Statement

Dr. Antonella Tosti is Editor-in-Chief of *Skin Appendage Disorders* Journal and acts as a consultant for Monat Global, Almirall, Tirthy Madison, Eli Lilly, Bristol Myers Squibb, P&G, Pfizer, and Myovant. The other authors have no conflict of interest to declare.

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Author Contributions

Rita Fernanda Cortez de Almeida prepared the draft, wrote the article, and approved the final version to be published. Daniel Fernandes Mello wrote and reviewed the article and approved the final version to be published. Antonella Tosti conceived the study, critically reviewed the article, and approved the final version to be published.